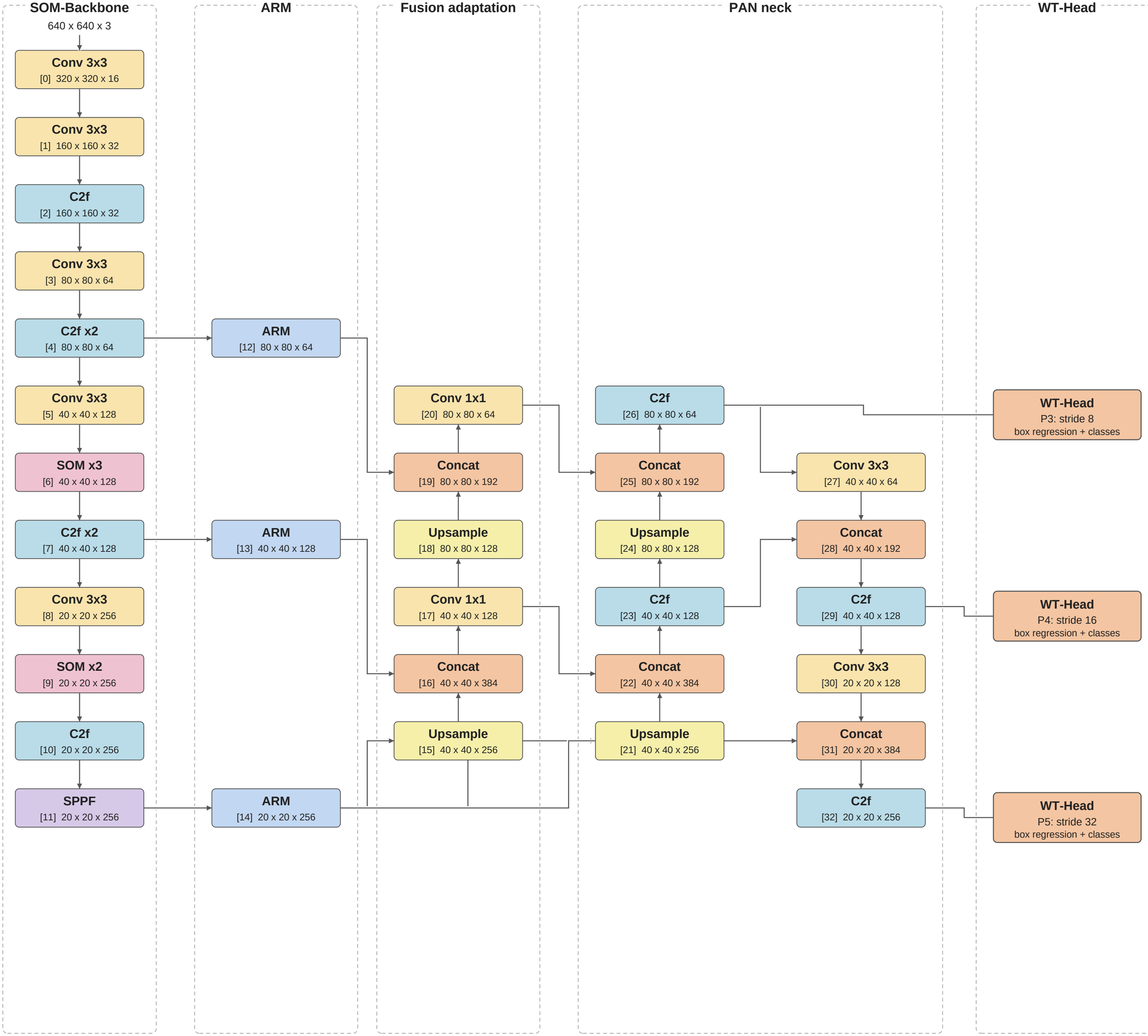
SOM-Backbone
640 x 640 x 3
Conv 3x3
[0] 320 x 320 x 16
Conv 3x3
[1] 160 x 160 x 32
C2f
[2] 160 x 160 x 32
Conv 3x3
[3] 80 x 80 x 64
C2f x2
[4] 80 x 80 x 64
Conv 3x3
[5] 40 x 40 x 128
SOM x3
[6] 40 x 40 x 128
C2f x2
[7] 40 x 40 x 128
Conv 3x3
[8] 20 x 20 x 256
SOM x2
[9] 20 x 20 x 256
C2f
[10] 20 x 20 x 256
SPPF
[11] 20 x 20 x 256
ARM
ARM
[12] 80 x 80 x 64
ARM
[13] 40 x 40 x 128
ARM
[14] 20 x 20 x 256
Fusion adaptation
Conv 1x1
[20] 80 x 80 x 64
Concat
[19] 80 x 80 x 192
Upsample
[18] 80 x 80 x 128
Conv 1x1
[17] 40 x 40 x 128
Concat
[16] 40 x 40 x 384
Upsample
[15] 40 x 40 x 256
PAN neck
C2f
[26] 80 x 80 x 64
Concat
[25] 80 x 80 x 192
Upsample
[24] 80 x 80 x 128
C2f
[23] 40 x 40 x 128
Concat
[22] 40 x 40 x 384
Upsample
[21] 40 x 40 x 256
Conv 3x3
[27] 40 x 40 x 64
Concat
[28] 40 x 40 x 192
C2f
[29] 40 x 40 x 128
Conv 3x3
[30] 20 x 20 x 128
Concat
[31] 20 x 20 x 384
C2f
[32] 20 x 20 x 256
WT-Head
WT-Head
P3: stride 8
box regression + classes
WT-Head
P4: stride 16
box regression + classes
WT-Head
P5: stride 32
box regression + classes